

YASH GHONGADE

Data Scientist | Machine Learning & Gen AI | RAG | LLMs | Python | Azure | AWS
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PROFESSIONAL SUMMARY

Software Developer skilled in Python, Machine Learning (prediction models), and AI projects, with hands-on experience in Machine Learning using Scikit-Learn Pipelines and Joblib. Proficient in NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, and with strong knowledge of Docker images, Docker containers, and Docker Hub along with that also have the knowledge of RAG and LLMs and MCP. Experienced in data collection techniques and practical Deep Learning concepts including Neural Networks. Also skilled in Web Development and eager to contribute and grow in the tech industry.

TECHNICAL SKILLS

Programming & Tools: Python, SQL, C, Java (Basics), Flask, Django, FastAPI, Web Development, Linux, Jupyter Notebook, Google Colab, Quadratic AI, Tableau

Data Science & Analytics: Pandas, NumPy, Scikit-Learn, Power BI, EDA, Feature Engineering, Matplotlib, Seaborn, Plotly

Machine Learning: Regression, Decision Trees, Random Forest, K-Means Clustering, Cross-Validation, RMSE, MAE, Scikit-Learn Pipelines, Feature Scaling, One-Hot Encoding, Stratified Shuffle Split.

Deep Learning: Neural Networks (ANN), TensorFlow, Keras, PyTorch, Model Fine-Tuning

Generative AI: Prompt Engineering, RAG, Agentic RAG, Re-ranking, Metadata Filtering, Embeddings, GPT, Claude, CodeX, Gemini, OpenAI API, Ollama, Local LLMs, FAISS, Chroma, Azure AI Search, AWS OpenSearch, Hugging Face, Semantic Search.

Agentic AI & Frameworks: LangChain, LangGraph, CrewAI, MCP, Multi-Agent Workflows, Tool & API Integration, Agent Orchestration

DevOps, Cloud & Databases: Docker, Terraform (IaC), Git, GitHub, GitLab, Kubernetes (Basics), CI/CD, MLOps, Azure, AWS (Bedrock, S3), Azure Blob Storage, MySQL, MongoDB

EXPERIENCE

Data Scientist Intern | Incubators Pvt. Ltd | *Jul 2025 – Feb 2026*

- Built end-to-end Machine Learning pipelines using Python, Scikit-Learn, Pandas, NumPy, including data preprocessing, feature engineering, and model training, with evaluation using RMSE, MAE.
- Performed Exploratory Data Analysis (EDA) using Matplotlib, Seaborn to extract actionable insights and improve data-driven decision making.
- Developed and deployed ML & Deep Learning models (Linear Regression, Random Forest, Decision Tree, TensorFlow, Keras) for prediction and classification tasks.
- Designed interactive data applications using Streamlit and worked on RAG-based systems using FAISS, Sentence Transformers, Chunking, Vectorization, LLMs, Prompt Engineering.
- Implemented data collection & processing pipelines using Web Scraping (BeautifulSoup, Requests), SQL (ETL, Joins, Window Functions, Joins, Trigger, View, Procedure), and ensured deployment with Docker, Git, GitHub.

Freelance AI/Data Engineer | Legal Data Intelligence — NY Court System (NYSCEF & WebCivil) | **Apr 2026 – Present**

Stack: Python · Selenium · Azure OCR · Azure RAG · BeautifulSoup · PyMuPDF · Pandas · HTML/JS · PyInstaller

- Engineered a dual-agent Producer–Consumer pipeline for automated large-scale PDF acquisition and structured data extraction from NYSCEF and WebCivil court portals, processing thousands of legal documents end-to-end.
- Built a robust web scraping system using Selenium, Crawl WebDriver, BeautifulSoup, and Requests with retry logic, session handling, and anti-bot resilience for uninterrupted court data collection at scale.
- Implemented an ETL pipeline (PyMuPDF + Pandas) integrated with Azure OCR & Azure RAG to extract key legal features from unstructured PDFs, transforming raw court filings into structured Excel/CSV datasets for a New York-based law firm.
- Packaged the entire scraping solution as a standalone Windows executable (.exe) using PyInstaller, enabling the client to run automated data collection independently without any Python environment setup.
- Designed a real-time monitoring dashboard (HTML/JavaScript) to track pipeline performance, scraping metrics, and failures, enabling proactive issue resolution.
- Conducted CAPTCHA analysis and bypass research — logged puzzle vs. image CAPTCHA frequency distributions and behavioral patterns in structured Excel reports; evaluated multiple solving strategies to maximise scraping success rate.
- Delivered client-facing export functionality (Excel + CSV download) enabling the law firm to directly consume and distribute extracted legal data across case workflows.

PROJECTS

RAG-Based AI Teaching Assistant (Video RAG) [RAG Project Link](#)

- Converted video/audio to text using OpenAI Whisper generated vector embeddings stored in FAISS for semantic similarity retrieval.
- Built a full RAG pipeline (retrieval + generation) integrating OpenAI API for hallucination-free, context-aware question answering with quality evaluation metrics.

MCP-Based AI Recruitment Screening & Shortlisting Assistant [MCP Project Link](#)

- Designed AI recruitment automation using Model Context Protocol - inspired modular architecture built Flask + MySQL backend with resume upload, PDF parsing (PyMuPDF), and skill-based candidate scoring.
- Integrated SMTP email automation for automated shortlist/rejection notifications. **Stack:** Python, Flask, MySQL, LLMs, Prompt Engineering, REST APIs.

Data Collection & Web Scraping Pipeline [Scraping Project Link](#)

- Scraped and extracted structured data from multiple websites using BeautifulSoup & Requests optimized workflow by caching HTML locally to reduce server hits and improve efficiency.

EDUCATION

B.E. Information Technology — SKNCOE, Pune | 2021–2025 | CGPA: **7.89**

ACHIEVEMENTS & CERTIFICATIONS

- **4th** Place – MIT Gen AI Hackathon
- **3rd** Place – Web Development Hackathon
- Data-Science
- Microsoft Azure AZ-900